

BATHULA SAI VARDHAN

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in Data Science, Machine Learning, and Generative AI. Proficient in Python, data analytics, LLM applications, Prompt Engineering, and Retrieval-Augmented Generation (RAG). Experienced in developing AI-powered solutions and data-driven applications, with a strong interest in leveraging modern GenAI frameworks to solve real-world problems. Passionate about continuous learning and emerging AI technologies.

WORK EXPERIENCE

AI Engineer Intern - Rudhra Info Solutions

June – Aug 2026 Electronic City, Bengaluru, India

- Built AI agents and an end-to-end **RAG** pipeline using **Gemini**, **Llama**, **LangChain**, and **ChromaDB** for intelligent document retrieval and conversational automation.
- Trained **Random Forest** and **XGBoost** models for predictive analytics and designed **Neo4j** knowledge graphs with **Cypher**-based graph retrieval integrated into AI workflows.
- Developed AI pipelines using **LlamaIndex**, **Scikit-learn**, **Pandas**, and **NumPy**; collaborated with senior engineers via **Git**-based workflows and code reviews.

Datamites - Data Science Intern

Jun 2025 - Dec 2025

- Worked on data analysis tasks and supported data organization for projects.
- Built classification models and created dashboards using Power BI. Analyzed datasets to identify trends and communicate insights through visualizations.

Towards Technology - Intern

Jul 2023 - Aug 2023

- Managed project timelines, reducing delivery times by 30%.
- Spearheaded the adoption of cutting-edge engineering software, improving project accuracy by 15%.
- Collaborated with cross-functional teams, enhancing project success rates by 10%.

PROJECTS

Clinical Protocol AI Agent | *Python, Gemini, LangChain, Qdrant, Streamlit*

- Built a **multi-agent RAG** pipeline using **Gemini 2.5 Flash**, **LangChain**, and **Qdrant** to automate clinical **Data Validation Plan (DVP)** generation with BAAI/bge-large-en-v1.5 embeddings.
- Engineered an **LLM-powered intent router** to dispatch prompts to a Q&A or structured generation engine, exporting Univariate, Multivariate, and Temporal edit checks to **.xlsx**.
- Developed a **Streamlit** app with **LlamaParse** PDF ingestion, **semantic chunking**, and **vector retrieval** for Q&A over clinical protocol documents.

AI Predictive Study Insights using Machine Learning

- Developed a clinical trial delay prediction app using **Python**, **XGBoost**, and **Streamlit** to forecast actual study end dates from synthetic trial data.
- Trained an **XGBoost Regression** model with feature engineering and One-Hot Encoding; evaluated using **MAE**, **RMSE**, and **R² Score**.
- Built an interactive **Streamlit** dashboard displaying predicted end date, delay days, risk level, and actionable insights.

EDUCATION

B.E in Computer science and Engineering

2021-2025

R.M.D Engineering College, Chennai | CGPA: 7.12

TECHNICAL SKILLS

- **Programming Languages:** Python, Java (Beginner), HTML
- **Data science:** Pandas, NumPy, Matplotlib, Seaborn, TensorFlow (Basic)
- **Generative AI:** Prompt Engineerin, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Retrieval-Augmented Generation (RAG), OpenAI API, Gemini API, AI Agents (Basic)
- **Machine Learning:** Linear Regression, Logistic Regression, Decision Tree
- **Tools:** Jupyter Notebook, Google Colab, Power BI, Tableau, VS Code, Antigravity, Kiro
- **Operating systems:** Windows, Linux, macOS